



November 24, 2025

a2z Radiology AI Inc.
Samir Rajpurkar
Chief Executive Officer
292 Newbury Street
Unit 235
Boston, MA 02115

Re: K252366

Trade/Device Name: a2z-Unified-Triage
Regulation Number: 21 CFR 892.2080
Regulation Name: Radiological Computer Aided Triage And Notification Software
Regulatory Class: Class II
Product Code: QAS, QFM,
Dated: November 1, 2025
Received: November 3, 2025

Dear Samir Rajpurkar:

We have reviewed your section 510(k) premarket notification of intent to market the device referenced above and have determined the device is substantially equivalent (for the indications for use stated in the enclosure) to legally marketed predicate devices marketed in interstate commerce prior to May 28, 1976, the enactment date of the Medical Device Amendments, or to devices that have been reclassified in accordance with the provisions of the Federal Food, Drug, and Cosmetic Act (the Act) that do not require approval of a premarket approval application (PMA). You may, therefore, market the device, subject to the general controls provisions of the Act. Although this letter refers to your product as a device, please be aware that some cleared products may instead be combination products. The 510(k) Premarket Notification Database available at <https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpmn/pmn.cfm> identifies combination product submissions. The general controls provisions of the Act include requirements for annual registration, listing of devices, good manufacturing practice, labeling, and prohibitions against misbranding and adulteration. Please note: CDRH does not evaluate information related to contract liability warranties. We remind you, however, that device labeling must be truthful and not misleading.

If your device is classified (see above) into either class II (Special Controls) or class III (PMA), it may be subject to additional controls. Existing major regulations affecting your device can be found in the Code of Federal Regulations, Title 21, Parts 800 to 898. In addition, FDA may publish further announcements concerning your device in the Federal Register.

FDA's substantial equivalence determination also included the review and clearance of your Predetermined Change Control Plan (PCCP). Under section 515C(b)(1) of the Act, a new premarket notification is not

required for a change to a device cleared under section 510(k) of the Act, if such change is consistent with an established PCCP granted pursuant to section 515C(b)(2) of the Act. Under 21 CFR 807.81(a)(3), a new premarket notification is required if there is a major change or modification in the intended use of a device, or if there is a change or modification in a device that could significantly affect the safety or effectiveness of the device, e.g., a significant change or modification in design, material, chemical composition, energy source, or manufacturing process. Accordingly, if deviations from the established PCCP result in a major change or modification in the intended use of the device, or result in a change or modification in the device that could significantly affect the safety or effectiveness of the device, then a new premarket notification would be required consistent with section 515C(b)(1) of the Act and 21 CFR 807.81(a)(3). Failure to submit such a premarket submission would constitute adulteration and misbranding under sections 501(f)(1)(B) and 502(o) of the Act, respectively.

Additional information about changes that may require a new premarket notification are provided in the FDA guidance documents entitled "Deciding When to Submit a 510(k) for a Change to an Existing Device" (<https://www.fda.gov/media/99812/download>) and "Deciding When to Submit a 510(k) for a Software Change to an Existing Device" (<https://www.fda.gov/media/99785/download>).

Your device is also subject to, among other requirements, the Quality System (QS) regulation (21 CFR Part 820), which includes, but is not limited to, 21 CFR 820.30, Design controls; 21 CFR 820.90, Nonconforming product; and 21 CFR 820.100, Corrective and preventive action. Please note that regardless of whether a change requires premarket review, the QS regulation requires device manufacturers to review and approve changes to device design and production (21 CFR 820.30 and 21 CFR 820.70) and document changes and approvals in the device master record (21 CFR 820.181).

Please be advised that FDA's issuance of a substantial equivalence determination does not mean that FDA has made a determination that your device complies with other requirements of the Act or any Federal statutes and regulations administered by other Federal agencies. You must comply with all the Act's requirements, including, but not limited to: registration and listing (21 CFR Part 807); labeling (21 CFR Part 801); medical device reporting (reporting of medical device-related adverse events) (21 CFR Part 803) for devices or postmarketing safety reporting (21 CFR Part 4, Subpart B) for combination products (see <https://www.fda.gov/combination-products/guidance-regulatory-information/postmarketing-safety-reporting-combination-products>); good manufacturing practice requirements as set forth in the quality systems (QS) regulation (21 CFR Part 820) for devices or current good manufacturing practices (21 CFR Part 4, Subpart A) for combination products; and, if applicable, the electronic product radiation control provisions (Sections 531-542 of the Act); 21 CFR Parts 1000-1050.

All medical devices, including Class I and unclassified devices and combination product device constituent parts are required to be in compliance with the final Unique Device Identification System rule ("UDI Rule"). The UDI Rule requires, among other things, that a device bear a unique device identifier (UDI) on its label and package (21 CFR 801.20(a)) unless an exception or alternative applies (21 CFR 801.20(b)) and that the dates on the device label be formatted in accordance with 21 CFR 801.18. The UDI Rule (21 CFR 830.300(a) and 830.320(b)) also requires that certain information be submitted to the Global Unique Device Identification Database (GUDID) (21 CFR Part 830 Subpart E). For additional information on these requirements, please see the UDI System webpage at <https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance/unique-device-identification-system-udi-system>.

Also, please note the regulation entitled, "Misbranding by reference to premarket notification" (21 CFR 807.97). For questions regarding the reporting of adverse events under the MDR regulation (21 CFR Part 803), please go to <https://www.fda.gov/medical-devices/medical-device-safety/medical-device-reporting-mdr-how-report-medical-device-problems>.

For comprehensive regulatory information about medical devices and radiation-emitting products, including information about labeling regulations, please see Device Advice (<https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance>) and CDRH Learn (<https://www.fda.gov/training-and-continuing-education/cdrh-learn>). Additionally, you may contact the Division of Industry and Consumer Education (DICE) to ask a question about a specific regulatory topic. See the DICE website (<https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance/contact-us-division-industry-and-consumer-education-dice>) for more information or contact DICE by email (DICE@fda.hhs.gov) or phone (1-800-638-2041 or 301-796-7100).

Sincerely,

A handwritten signature in black ink that reads "Jessica Lamb". The signature is written in a cursive style. In the background, there is a large, light blue watermark of the letters "FDA".

Jessica Lamb, PhD
Assistant Director
Imaging Software Team
DHT8B: Division of Radiological Imaging
Devices and Electronic Products
OHT8: Office of Radiological Health
Office of Product Evaluation and Quality
Center for Devices and Radiological Health

Enclosure

Indications for Use

Please type in the marketing application/submission number, if it is known. This textbox will be left blank for original applications/submissions.

K252366

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Please provide the device trade name(s).

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a2z-Unified-Triage

Please provide your Indications for Use below.

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a2z-Unified-Triage is a radiological computer-aided triage and notification software indicated for use in the analysis of abdominal/pelvic CT images in adults aged 22 and older. The device is intended to assist hospital networks and appropriately trained medical specialists in workflow triage by flagging and communicating suspected positive cases of the 7 specified abdominopelvic findings: Acute Cholecystitis, Acute Pancreatitis, Unruptured Abdominal Aortic Aneurysm, Acute Diverticulitis, Free Air, Hydronephrosis, and Small Bowel Obstruction. These findings are intended to be used together as one device. The device supports both cloud-based and on-premises deployment, with integration either directly with healthcare facility systems or through third-party healthcare technology platforms.

a2z-Unified-Triage uses an artificial intelligence algorithm to analyze images and flag cases with detected findings in parallel to the ongoing standard of care image interpretation. The device provides analysis results that enable client systems to generate notifications for cases with suspected findings. These results can include DICOM instance UIDs for key images, which are meant for informational purposes only and not intended for primary diagnosis beyond notification. The device does not alter the original medical image and is not intended to be used as a diagnostic device.

The results of a2z-Unified-Triage are intended to be used in conjunction with other patient information and based on clinicians' professional judgment, to assist with triage/prioritization of medical images. Notified clinicians are responsible for viewing full images per the standard of care.

Please select the types of uses (select one or both, as applicable).

- ☒ Prescription Use (Part 21 CFR 801 Subpart D)
☐ Over-The-Counter Use (21 CFR 801 Subpart C)

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510(k) Summary

1. Administrative Information

Submitter Name	a2z Radiology AI Inc.
Address	292 Newbury Street, Unit 235, Boston, MA 02115, USA
Phone Number	+1 508-293-1822
Fax Number	N/A
Company Representative	Samir Rajpurkar
Email	support@a2zradiology.ai
Date Summary Prepared	November 01, 2025

2. Subject Device Information

Trade Name	a2z-Unified-Triage
Subject Device K Number	K252366
Common Name	Radiological computer aided triage and notification software
Product Code	QAS, QFM
Regulation Number	892.2080
Regulatory Class	Class II
Review Panel	Radiology

3. Predicate Device Information

Predicate Device Name	Annalise Enterprise CTB Triage Trauma
Predicate Device K Number	K223240
Common Name	Radiological computer aided triage and notification software
Product Code	QAS
Regulation Number	892.2080
Regulatory Class	Class II
Review Panel	Radiology

This predicate has not been subject to a design-related recall. No reference devices were used in this submission.

4. Device Description

a2z-Unified-Triage is a radiological computer-assisted triage and notification software device. The software consists of an algorithmic component that supports both cloud-based and on-premises

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deployment on standard server hardware. The device processes abdomen/pelvis CT images from clinical imaging systems, analyzing them using artificial intelligence algorithms to detect suspected cases of 7 abdominopelvic conditions: Acute Cholecystitis, Acute Pancreatitis, Unruptured Abdominal Aortic Aneurysm, Acute Diverticulitis, Free Air, Hydronephrosis, and Small Bowel Obstruction.

Following the AI processing, the analysis results are returned to the client system for worklist prioritization. When a suspected case is detected, the software provides analysis results that enable the client system to generate appropriate notifications. These results can include DICOM instance UIDs for key images, which are for informational purposes only, do not contain any marking of the findings, and are not intended for primary diagnosis beyond notification.

Integration with clinical imaging systems facilitates efficient triage by enabling prioritization of suspect cases for review of the relevant original images in the PACS. Thus, the suspect case receives attention earlier than would have been the case in the standard of care practice alone.

Algorithm Architecture: The artificial intelligence algorithms implement deep learning models that analyze CT studies and return binary outputs indicating the presence or absence of each of the 7 target abdominopelvic findings. When a finding is detected, the algorithm identifies a key image slice most likely to show the condition, provided for informational purposes only.

Training Database: The algorithms were developed on an extensive dataset of abdomen/pelvis CT studies from multiple clinical sites. The training data encompassed diverse imaging characteristics including multiple CT scanner manufacturers, both contrast-enhanced and non-contrast studies, variable slice thicknesses, and a wide range of patient demographics.

Data Independence: Strict protocols ensured complete independence between development and testing datasets, with mutually exclusive patients. No overlap existed between training/validation data and the independent test dataset used for performance evaluation.

5. Indications for Use

a2z-Unified-Triage is a radiological computer-aided triage and notification software indicated for use in the analysis of abdominal/pelvic CT images in adults aged 22 and older. The device is intended to assist hospital networks and appropriately trained medical specialists in workflow triage by flagging and communicating suspected positive cases of the 7 specified abdominopelvic findings: Acute Cholecystitis, Acute Pancreatitis, Unruptured Abdominal Aortic Aneurysm, Acute Diverticulitis, Free Air, Hydronephrosis, and Small Bowel Obstruction. These findings are intended to be used together as one device. The device supports both cloud-based and on-premises deployment, with integration either directly with healthcare facility systems or through third-party healthcare technology platforms.

a2z-Unified-Triage uses an artificial intelligence algorithm to analyze images and flag cases with detected findings in parallel to the ongoing standard of care image interpretation. The device provides analysis results that enable client systems to generate notifications for cases with suspected findings. These results can include DICOM instance UIDs for key images, which are meant for informational purposes only and not intended for primary diagnosis beyond notification. The

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device does not alter the original medical image and is not intended to be used as a diagnostic device.

The results of a2z-Unified-Triage are intended to be used in conjunction with other patient information and based on clinicians' professional judgment, to assist with triage/prioritization of medical images. Notified clinicians are responsible for viewing full images per the standard of care.

Comparison of Technological Characteristics

The subject device, a2z-Unified-Triage, is substantially equivalent to the primary predicate Annalise Enterprise CTB Triage Trauma (K223240).

The subject and predicate are both radiological computer-aided triage and notification software using artificial intelligence algorithms. Both aid radiological image triage through deep learning algorithms trained on medical images, providing specialists with notifications and preview images for preemptive triage. Neither removes cases from standard care reading queues nor de-prioritizes cases. Both operate in parallel to standard care as the default option.

While the anatomical regions of interest differ (abdomen/pelvis for the subject device vs. head for the predicate), the core technology, intended use for triage, and principles of operation are similar. Both devices raise identical safety and effectiveness questions regarding accurate triage. Performance testing, summarized below, demonstrated that the subject device maintains equivalent safety and effectiveness for its intended use. A detailed comparison of key features is provided in the table below.

6. Comparison of Technological Characteristics with the Predicate Device

Table 1. Comparison with predicate device.

Characteristic	Subject Device: a2z-Unified-Triage	Predicate Device: Annalise Enterprise CTB Triage Trauma (K223240)
Manufacturer	a2z Radiology AI Inc.	Annalise-AI Pty Ltd
Regulation Number	892.2080	892.2080
Regulatory Class	Class II	Class II
Product Code	QAS, QFM	QAS
Regulation Name	Radiological computer aided triage and notification software	Radiological computer aided triage and notification software
Device Property	SaMD (Software as a Medical Device)	SaMD (Software as a Medical Device)

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Characteristic	Subject Device: a2z-Unified-Triage	Predicate Device: Annalise Enterprise CTB Triage Trauma (K223240)
Indications for Use	Radiological computer-aided triage and notification software for analysis of abdominal/pelvic CT images in adults aged 22+. Assists in workflow triage by flagging 7 abdominopelvic findings.	Device to aid in triage and prioritization of studies with features suggestive of 4 intracranial hemorrhage findings in patients 22+.
Technical Characteristics		
Input	Abdomen/Pelvis CT images (with or without contrast)	Non-contrast brain CT images
Output	JSON object with binary classification for 7 findings and DICOM instance UUIDs for positive cases	Notification to image and order management system for worklist prioritization
Algorithm Type	Non-adaptive machine learning (Deep Learning - Neural Networks)	Non-adaptive machine learning (Deep Learning - Neural Networks)
Intended Users	Appropriately trained medical specialists	Trained clinicians qualified to interpret brain CT studies
Target Population	Adults (22 years and older)	Adults (22 years and older)
Location of anatomical structures	Abdomen and Pelvis	Head (Brain)
Imaging Modality	Computed Tomography (CT)	Computed Tomography (CT)
Intended Use Environment	Hospital environment or other clinical settings with DICOM-compliant CT imaging and IT infrastructure	Medical care environment
Performance	7 findings met targets: QAS findings >80% sensitivity/specificity; QFM findings >0.95 AUC	Met performance targets with sensitivity and specificity >80% across findings
Triage Turn-around Time	Mean: 58.39 seconds (95% CI: 56.11-60.68), Median: 55.02 seconds, 95th percentile: 90.36 seconds	Mean: 81.6 seconds (95% CI: 80.3-82.9)

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Characteristic	Subject Device: a2z-Unified-Triage	Predicate Device: Annalise Enterprise CTB Triage Trauma (K223240)
Software device that operates on off-the-shelf hardware	Yes. Software supports both cloud-based and on-premises deployment on standard server hardware	Yes. Interfaces with image and order management systems

7. Performance Data

7.1. Software Verification and Validation Testing

Software verification and validation testing were conducted, and documentation was provided as recommended by FDA’s Guidance for Industry and FDA Staff, “Guidance for the Content of Premarket Submissions for Device Software Functions”.

7.2. Performance Assessment

A standalone performance assessment was performed for a2z-Unified-Triage to validate the accuracy of detecting the 7 findings against a reference standard established by U.S. board-certified radiologists.

7.2.1. Ground Truth The ground truth for the 678-case Test Data was established using a 2+1 methodology exclusively with U.S. board-certified radiologists. Specifically, each case was independently reviewed by two U.S. board-certified radiologists. For cases where the two readers disagreed (discordant cases), a third U.S. board-certified expert adjudicator, fellowship-trained in body imaging, provided the tiebreaking determination to establish the final ground truth. This ensures that all ground truth determinations are made by qualified U.S. board-certified radiologists using clinical standards consistent with U.S. practice patterns.

7.2.2. Development and Test Dataset a2z-Unified-Triage was trained and tested on an extensive and diverse set of abdominal/pelvic CT studies from multiple clinical sites. The final test cohort consisted of 678 cases, with 3 cases excluded due to quality control failures during model inference, resulting in an analytic cohort of 675 cases from 643 unique patients. Development and test cohorts had mutually exclusive patients. The test dataset was constructed to ensure adequate representation for each of the findings and included a wide range of patient demographics and imaging characteristics, as summarized below.

Test Cohort Demographics and Imaging Characteristics (N=675)

Sex	N	Percentage
Female	345	51.1

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Sex	N	Percentage
Male	330	48.9

Age (years): Median (IQR) = 61 (44-75); Mean (SD) = 59.9 (18.2); Range = 22-90

Site and State Distribution

Site	State	Number of Cases
Site 1	New York	223 (33.0%)
Site 2	Kansas	120 (17.8%)
Site 3	New York	82 (12.1%)
Site 4	Missouri	66 (9.8%)
Site 5	Texas	61 (9.0%)
Other (7 Sites)	Various	123 (18.2%)
Total		675

Sites in 'Other' category were located in Missouri, Texas, Kansas, and Nebraska. Total unique sites: 12

State Distribution:

State	N	Percentage
New York	305	45.2
Kansas	143	21.2
Missouri	124	18.4
Texas	101	15.0
Nebraska	2	0.3

7.3.3. Imaging Characteristics Manufacturer Distribution:

Manufacturer	N	Percentage
GE	277	41.0
SIEMENS	217	32.1
Canon	107	15.9
TOSHIBA	52	7.7
Other	22	3.3

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Manufacturers in ‘Other’ category: FUJI, Hitachi, Philips

Contrast Status:

Contrast Status	N	Percentage
With Contrast	385	57
Without Contrast	290	43

Slice Thickness:

Slice Thickness	N	Percentage
5mm	464	69
2mm to <5mm	211	31

Mean (SD): 4.30 (1.06) mm; Median (IQR): 5.00 (3.00-5.00) mm; Range: 2-5mm

7.2.4. Clinical Subgroups and Confounders The test dataset included diverse disease presentations across all 7 target conditions. U.S. board-certified radiologists annotated clinically relevant anatomical variations and disease subtypes on subsets of positive cases, including: calculous and acalculous acute cholecystitis, gallbladder wall thickening, and gallbladder hydrops; acute pancreatitis with and without fluid collections; sigmoid and other colonic locations of acute diverticulitis including cases with perforation; fusiform and saccular unruptured AAA morphologies across multiple size categories and anatomical locations; varying severities of hydronephrosis; small bowel obstruction from adhesive and mechanical etiologies; and free air from various etiologies and volumes. The test cohort also included cases with concurrent findings. Comprehensive stratified analyses demonstrated consistent device performance across all demographic subgroups, imaging parameters, clinical sites, and disease presentations.

7.2.5. Primary Endpoints

Condition	Positive Cases	Total Cases	Prevalence (%)
Acute Cholecystitis	77	675	11.4%
Acute Pancreatitis	99	675	14.7%
Unruptured AAA	76	675	11.3%
Acute Diverticulitis	76	675	11.3%
Free Air	112	675	16.6%
Hydronephrosis	97	675	14.4%
Small Bowel Obstruction	99	675	14.7%

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7.2.4. QFM Conditions

Condition	Available Operating Points and Performance
Acute Cholecystitis (QFM) Total N=675, Positive=77 AUC: 0.985 [0.972-0.998]	High Sensitivity: Se 96.1% [89.2-98.7%], Sp 89.3% [86.6-91.5%] Sensitivity Biased: Se 92.2% [84.0-96.4%], Sp 95.8% [93.9-97.2%] Balanced: Se 92.2% [84.0-96.4%], Sp 95.8% [93.9-97.2%]
Acute Pancreatitis (QFM) Total N=675, Positive=99 AUC: 0.994 [0.985-1.000]	High Sensitivity: Se 98.0% [92.9-99.4%], Sp 87.8% [84.9-90.3%] Sensitivity Biased: Se 98.0% [92.9-99.4%], Sp 97.0% [95.3-98.1%] Balanced: Se 98.0% [92.9-99.4%], Sp 97.0% [95.3-98.1%] High Specificity: Se 92.9% [86.1-96.5%], Sp 99.8% [99.0-100.0%]
Acute Diverticulitis (QFM) Total N=675, Positive=76 AUC: 0.995 [0.990-1.000]	High Sensitivity: Se 98.7% [92.9-99.8%], Sp 89.3% [86.6-91.5%] Sensitivity Biased: Se 97.4% [90.9-99.3%], Sp 96.8% [95.1-98.0%] Balanced: Se 97.4% [90.9-99.3%], Sp 96.8% [95.1-98.0%] High Specificity: Se 94.7% [87.2-97.9%], Sp 98.7% [97.4-99.3%]
Unruptured AAA (QFM) Total N=675, Positive=76 AUC: 0.995 [0.991-0.999]	High Sensitivity: Se 100.0% [95.2-100.0%], Sp 86.3% [83.3-88.8%] Sensitivity Biased: Se 97.4% [90.9-99.3%], Sp 95.8% [93.9-97.2%] Balanced: Se 97.4% [90.9-99.3%], Sp 97.5% [95.9-98.5%]
Hydronephrosis (QFM) Total N=675, Positive=97 AUC: 0.976 [0.960-0.991]	High Sensitivity: Se 89.7% [82.1-94.3%], Sp 92.9% [90.5-94.7%]

7.2.5. QAS Conditions

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Condition	Available Operating Points and Performance
Small Bowel Obstruction (QAS) Total N=675, Positive=99	High Sensitivity: Se 94.9% [88.7-97.8%], Sp 91.7% [89.1-93.7%] Sensitivity Biased: Se 91.9% [84.9-95.8%], Sp 96.0% [94.1-97.3%] Balanced: Se 88.9% [81.2-93.7%], Sp 98.1% [96.6-98.9%]
Free Air (QAS) Total N=675, Positive=112	Balanced: Se 89.3% [82.2-93.8%], Sp 88.6% [85.7-91.0%] High Specificity: Se 88.4% [81.1-93.1%], Sp 90.8% [88.1-92.9%]

7.2.6. Performance Discussion The comprehensive performance evaluation of a2z-Unified-Triage demonstrates consistently exceptional performance across all seven abdominopelvic findings, meeting or substantially exceeding FDA-specified acceptance criteria. All QAS findings (Free Air, Small Bowel Obstruction) met the acceptance criteria of sensitivity and specificity >80%, while 5 QFM findings (Acute Cholecystitis, Acute Pancreatitis, Unruptured Abdominal Aortic Aneurysm, Acute Diverticulitis, Hydronephrosis) met the acceptance criteria of AUC >0.95.

The extensive stratified analyses across patient demographics (sex, age, site, state), imaging parameters (contrast status, slice thickness, equipment manufacturer), and clinical scenarios (concurrent findings) reveal robust performance maintenance with no clinically significant degradation across any subgroup. Additionally, comprehensive stratified analyses by anatomical variations and edge cases were performed for all seven conditions, with U.S. board-certified radiologists annotating clinically relevant attributes on subsets of positive cases. These analyses—including disease subtypes, confounders, and morphological features specific to each condition—demonstrate robust device performance across all clinically relevant variations, validating that the device independently and accurately detects each indication based on its primary imaging features rather than confounding concurrent findings. This validates the algorithm’s generalizability and reliability across diverse clinical presentations. The triage effectiveness study confirms rapid processing times well within clinical workflow requirements, establishing a2z-Unified-Triage as a clinically viable solution for enhancing radiological workflow efficiency.

8. Predetermined Change Control Plan (PCCP)

This submission contains a Predetermined Change Control Plan (PCCP). The PCCP does not include provisions for implementation of adaptive algorithms that will continuously learn in the field. All algorithm modifications will be trained, validated, and locked prior to release of the software to the field. A procedure has been established for updating device labeling to inform users about changes implemented under this PCCP, including a summary of the changes, characterization of algorithm performance, and compatibility information.

The PCCP specifies possible modifications to a2z-Unified-Triage, as well as verification and validation activities to implement changes in a controlled manner such that the modified device remains as safe and effective as the originally cleared device. The PCCP applies to the seven conditions val-

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idated in this 510(k) submission: Acute Cholecystitis, Acute Pancreatitis, Unruptured Abdominal Aortic Aneurysm, Acute Diverticulitis, Free Air, Hydronephrosis, and Small Bowel Obstruction. For each condition, the PCCP covers all operating points that meet pre-specified performance criteria upon revalidation.

The planned modifications include:

- Training data volume and diversity expansion
- Training data quality and annotation refinement
- Ensemble composition optimization
- Data augmentation parameter adjustments
- Training parameter optimization
- Weight initialization method selection
- Neural network architecture component modifications
- Input data preprocessing adjustments
- Validated slice thickness range expansion

All modifications implement bounded specifications with specific parameter ranges to ensure changes remain within validated frameworks. Comprehensive validation requirements apply to all modifications, including minimum test dataset size, ground truth determination by U.S. board-certified radiologists, complete data sequestration, stratified subgroup analysis, and acceptance criteria requiring sensitivity and specificity >80% for all findings plus AUC >0.95 for QFM findings.

The modification protocol incorporates impact assessment considerations and specifies requirements for data management, including data sources, collection, storage, and sequestration, as well as documentation and data re-use practices. Detailed validation activities, testing methodologies, and performance requirements have been established for each modification. All changes will undergo appropriate verification and validation testing before implementation to ensure the modified device maintains the safety and effectiveness profile of the originally cleared device.

9. Conclusion

The subject device, a2z-Unified-Triage, and the predicate, Annalise Enterprise CTB Triage Trauma, are substantially equivalent. Both are software devices intended to aid in the prioritization and triage of radiological images using AI algorithms. They share the same fundamental scientific technology and principles of operation.

While the specific anatomical regions and findings differ, the intended use for triage, the AI-based approach, the integration into clinical workflows, and the parallel nature of operation are highly similar. Both devices aim to reduce turnaround time through preemptive triage without altering the standard of care.

The performance data for a2z-Unified-Triage demonstrates that the device is safe and effective for its intended use and performs as intended. Therefore, a2z-Unified-Triage is substantially equivalent to the predicate device.